

Binary event perpetuals: corrected theory, risk mathematics, and contract requirements

Review date: 29 September 2026. **Scope:** the mechanism motivated by [paper1.pdf](#), with an adversarial audit of the attached *Perps on Binary Outcomes - Theory and Mechanism Design*. The larger Metropolis plan is not the product specification for this analysis.

Verdict: a solvent leveraged event market is possible if its contingent obligations are actually funded, collectible, hedged, or contractually reduced. The draft does **not** prove its stronger claim that its proposed margin schedule, funding, and pre-resolution reductions guarantee zero bad debt. Several intermediate claims are false. The missing step is acquiring and preserving the required backing before an adverse information jump, including when nobody can execute a liquidation.

This report supplies corrected proofs, counterexamples, an implementable accounting model, explicit capital constraints, and the additional evidence required before those constraints can support production contracts. It is a mathematical and engineering specification, not a claim that unprovided contracts or unprovided historical data have been audited. Statements are labeled **identity/proof**, **conditional model**, **policy**, or **unverified empirical input** where that distinction matters.

Read this together with [contract invariants and tests](#), [risk validation plan](#), and [reproducibility notes](#). The companion files specify transitions, measurements, and acceptance gates in more detail.

Contents

1. What is being solved	3
2. Fix the economic contract and notation first	3
3. The ledger that should underlie the contracts	4
4. What cannot be obtained for free, and what can be built	5
5. Probability theory: the valid theorem and its limits	7
6. Correct tail measures and a usable margin framework	8
7. Liquidation and compression require an executable mechanism	10
8. Open orders are contingent obligations	12
9. Funding must preserve solvency as well as sum to zero	13
10. Settlement and conversion	15
11. Oracle, mark, manipulation, and collateral risk	16
12. Repairing the draft's worked example	19
13. Claim-by-claim verdict on the attached draft	20
14. The actual safety theorem to implement	21
15. What risk analysis must measure	22
16. Evidence delivered and limits	23
17. Source record	24

1. What is being solved

The original paper is explicitly a failure-mode study and mechanical replay, not a validated exchange design. Its entry-to-terminal balance-sheet result survives this review. Its negative replay results do not establish that the proposed controls are deployable, and do not prove every possible leveraged design impossible. We have not reproduced its historical run: the attached PDFs do not supply the full executable data and experiment environment.

The draft adds useful ideas, especially bounded terminal losses and conversion into funded outcome tokens. But it sometimes promotes an accounting identity into an execution guarantee, or a statistical model into a theorem. Those promotions must be removed before writing contracts.

Four different questions must have separate answers:

1. **What is owed?** The signed cash and position ledger determines this exactly.
2. **Who supplies missing capital?** Traders, an underwriting pool, secured recourse, a hedge, or claimants through disclosed loss sharing.
3. **How often and how badly will the capital be used?** This needs a calibrated joint model of information, order flow, liquidity, latency, and behavior.
4. **Does the implementation enforce its promises?** This needs transition proofs, integer arithmetic tests, adversarial execution tests, and an actual contract audit.

“We intend to liquidate before resolution” is an operating strategy. It is not an answer to question 2.

2. Fix the economic contract and notation first

Use one specified collateral token as the numeraire. A normal event outcome is $Y \in \{0, 1\}$; one claim pays one collateral unit for YES and zero for NO. If invalid events pay a fraction, specify the permitted $y \in [0, 1]$ and its determination before listing. A collateral unit need not retain one US dollar of purchasing power.

Symbol	Meaning	Unit / restriction
p_t	$\mathbb{P}(Y = 1 \mid \mathcal{F}_t)$, conditional outcome probability	Dimensionless; latent and model dependent
S_t	Executable spot price of the identical claim	Collateral per claim
I_t	Published index	Collateral per claim; not automatically executable
m_t	Contractual risk mark	Collateral per claim
P	Actual fill price	Collateral per claim
z_i	Signed claim quantity; positive is long YES	Claims
b_i	Signed cash intercept after economic accruals	Collateral
$E_i(m)$	Account equity at mark m	Collateral
F	Cumulative funding index	Collateral per claim
R	Available, unencumbered loss reserve	Collateral; excludes senior liabilities
Q	Open interest, counted on one side	Claims
ϵ	Statistical tail probability budget	In $(0, 1)$; not a solvency guarantee
H	Actual closeout horizon	Random time, dependent on position and market state

Keep probability, index, mark, and execution price separate in storage, APIs, simulations, and papers. In particular, neither a clamped mark nor a spot quote becomes the physical conditional probability by definition.

Under a pricing measure $\mathbb{Q}$, deterministic rate r , and fixed payment time T , a frictionless pricing expression is $S_t = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_t]$. With random payment time, discount inside the expectation. Stablecoin collateral does not imply zero interest, zero credit risk, or equality of physical and pricing probabilities. The accounting proofs below need none of these pricing assumptions.

3. The ledger that should underlie the contracts

3.1 Exact state and transitions

Identity A1:

$$E_i(m) = b_i + z_i m, \quad E_i(0) = b_i, \quad E_i(1) = b_i + z_i. \quad (\text{A1})$$

For a signed fill δ at P , after funding on the old position is accrued:

$$z'_i = z_i + \delta, \quad b'_i = b_i - \delta P - f_i. \quad (\text{A2})$$

The counterparty receives the opposite position and the same cash amount; fees transfer into an explicit fee ledger. Deposits increase b ; withdrawals decrease it. Funding over an unchanged position changes b_i by $-z_i \Delta F$. These are economic changes even if physical token transfers stay inside a vault.

Average entry price is a display and realized-P&L convention. For a fixed position it is possible to write $b = C - ze$, with C including all realized cash, fees, and funding. On a sign flip, close the old position and open the remainder. Do not maintain two independent “truths” in entry-price accounting and cash accounting.

For a matched market:

$$\begin{aligned} \sum_i z_i &= 0, & Q &= \sum_i z_i^+ = \sum_i (-z_i)^+, \\ T &= \sum_i b_i = \sum_i E_i(m). \end{aligned} \quad (\text{A3})$$

Here $x^+ = \max(x, 0)$. Vault reconciliation must include every category: trader signed ledger T , reserve, fees, free balances outside this ledger, token inventories, and rounding dust, without double counting. A negative account is a claim against that account, not spendable collateral.

Crucial distinction: signed conservation $\sum E_i = T$ does not imply that $\sum E_i^+$ can be paid after negative accounts default. The following reserve derivation closes that gap.

3.2 Endpoint backing

Worst price loss from mark $m \in [0, 1]$ is

$$W_i(m) = z_i^+ m + (-z_i)^+ (1 - m). \quad (\text{A4})$$

Subtracting gives

$$E_i(m) - W_i(m) = b_i + \min(z_i, 0) = \min\{b_i, b_i + z_i\}. \quad (\text{A5})$$

Theorem A: with this linear payoff and no additional debits, an account covers both outcomes if and only if

$$b_i \geq 0, \quad b_i + z_i \geq 0. \quad (\text{A6})$$

Proof: these are the two possible terminal equities. Equation A5 shows the equivalent mark-based test. This condition is independent of the mark; changing the mark cannot create backing. Actual trades, withdrawals, fees, and funding can change it.

Future fees, committed orders, and authorized funding need additional slack. Define $S_i = \min(b_i, b_i + z_i)$. Requirements on those future debits are requirements on S_i , not just on marked equity.

Define displayed leverage $L = W/E$ only when its denominator and interpretation are valid. For $E \leq 0$, report insolvency explicitly. For flat or zero-risk positions, use explicit branches. Contract health checks should compare integers, not divide by equity. With future debit reserve $K > 0$, full backing requires $E \geq W + K$, so the associated leverage can be below one.

3.3 Exact reserve for limited-liability accounts

Suppose positive terminal equity is paid in full and negative terminal equity is forgiven. For each endpoint define

$$D_y = \sum_i [-(b_i + z_i y)]^+, \quad P_y = \sum_i [b_i + z_i y]^+. \quad (\text{A7})$$

Because positive part minus negative part equals the original number, A3 implies

$$P_y = T + D_y. \quad (\text{A8})$$

Theorem B: if liquid collateral assets available to these claims are $T + R$, the exact additional cash reserve needed to pay all positive claims under either normal outcome is

$$R_* = \max(D_0, D_1). \quad (\text{A9})$$

Proof: $T + R \geq P_y$ is equivalent to $R \geq D_y$ for each endpoint. Necessity and sufficiency follow. This is a settlement coverage theorem, conditional on assets being available and the ledger being correct; it does not make underwriters profitable.

A simpler but more conservative reserve is $R_{\text{sep}} = \sum_i \max(d_{i0}, d_{i1})$, where $d_{iy} = (-E_i(y))^+$. Then

$$R_* \leq R_{\text{sep}} \leq D_0 + D_1 \leq 2R_*. \quad (\text{A10})$$

Pooling can therefore reduce required cash for one binary event. It does not permit treating every account default as independent: all accounts share the same outcome.

For fractional invalid payout $y \in [0, 1]$, $D(y)$ is convex, as a sum of positive parts of affine functions. Therefore $D(y) \leq \max(D(0), D(1))$. Endpoint cash coverage also covers every such fraction, provided the payoff really is the same linear function and costs do not change by outcome.

If the reserve holds an identical secured hedge with payoff H_y , replace A9 with

$$R_{\text{cash}} + H_y \geq D_y \quad \text{for every allowed } y. \quad (\text{A11})$$

Paying for the hedge reduces reserve cash. Redemption timing, custody, collateral, oracle, and chain must match the liability. A token redeemable later on another chain is not necessarily liquid cash for an immediate withdrawal.

For several events, use $E_i(\mathbf{y}) = b_i + \sum_j z_{ij} y_j$ and test every contractually possible joint scenario. Correlation estimates do not justify excluding logically possible states from a worst-case promise. Cross-event reserve optimization can be a discrete optimization problem; a one-event two-endpoint check is no longer sufficient.

4. What cannot be obtained for free, and what can be built

4.1 The jump obstruction

Take a long n at entry price e , collateral $C < ne$, with no intervening top-up, close, external payment, or payoff change. At NO:

$$E_0 = C - ne < 0. \quad (\text{B1})$$

If $C = ne/L$, the shortfall is $ne(1 - 1/L)$. A pre-resolution mark m cancels:

$$C + n(m - e) + n(0 - m) = C - ne.$$

This proves a precise impossibility: under these assumptions, no mark formula, monitoring speed, or liquidation trigger can guarantee nonnegative terminal equity if adverse revelation may precede executable action. It does not say every outcome defaults, nor that a funded exchange cannot offer leverage.

4.2 Three coherent product modes

Mode	Promise and hard admission invariant	Role of statistical margin
Individually backed	A6, plus order/funding/fee reserves for every account	Optional operating control; no missing terminal price capital
Underwritten leverage	Traders may violate A6; secured pool satisfies A9/A11 after every admitted change	Determines trader margin, premiums, position limits, and risk reduction attempts
Explicit loss-bearing leverage	Finite assets can be smaller than full promises; contractual ADL/haircut/default terms specify reduced payments	Estimates frequency and severity; UI cannot promise unconditional full payout

Secured recourse is another source of assets only to the extent it is genuinely enforceable and collectible on the needed schedule. An unsecured promise to top up is not present collateral.

My proposed direction for the leveraged version is **finite, funded underwriting with exact endpoint admission controls**, accompanied by an explicit emergency loss-allocation rule for failures outside the stated model. This directly models reality: it exposes how much capital leverage consumes, when new trades must be rejected, and when the business is uneconomic. It does not silently refill a reserve or substitute testnet sponsorship for a real risk model.

4.3 Why free default protection is exploitable

At price .60, match a 1,000-unit long with margin 120 and short with margin 80. Then:

Outcome	Long signed equity	Short signed equity	Payable claims after forgiveness	Deficit
NO	-480	680	680	480
YES	520	-320	520	320

Total deposits are 200, but positive claims are at least 520 in either outcome. A single actor controlling both wallets obtains a guaranteed subsidy if the reserve covers defaults without an adequate charge or other constraint. The exact pooled reserve is 480; summing individual worst deficits gives 800.

Under frictionless identical state prices .60/.40, the claim to these reserve payments costs $.40(480) + .60(320) = 384$. This is a replication benchmark, not a real underwriting premium recommendation. Adverse selection, funding, capital costs, execution, and claim timing alter real pricing.

4.4 The hedge / payoff equivalence

For a fixed non-recourse long with $C < ne$, forgiven negative balances, and no uncertain future charges:

$$[C + n(Y - e)]^+ = [C + n(1 - e)]Y. \quad (B2)$$

It is a fixed number of YES claims. Equivalently, buying $D = ne - C$ NO claims covers the unprotected position's adverse deficit. If YES costs e , NO costs $1 - e$, and complete-set parity is executable, total cost is

$$C + (1 - e)(ne - C) = e[C + n(1 - e)]. \quad (B3)$$

The protected exposure costs its spot replication value. This does not mean leverage has no economic use: intraday financing, capital allocation, hedged portfolios, and separately priced insurance can be useful. It means unpaid tail losses do not disappear. Future uncertain funding makes the payoff path dependent, so B2 is then not a fixed-token replication.

For any fixed current book with identical traded state claims, the theoretical hedge cost of terminal reserve payouts is $(1 - S)D_0 + SD_1$, subject to execution and matching assumptions. Worst-case cash $\max(D_0, D_1)$, expected physical loss, and hedge cost are three different quantities.

5. Probability theory: the valid theorem and its limits

5.1 The conditional variance identity

Let $p_t = \mathbb{E}[Y \mid \mathcal{F}_t]$, and let τ reveal Y . For a finite increasing, nonanticipating partition $t = t_0 \leq \dots \leq t_N = \tau$, under optional-sampling conditions (bounded stopping times suffice), conditional martingale increments are orthogonal. Thus

$$\begin{aligned} \mathbb{E} \left[\sum_{k=1}^N (p_{t_k} - p_{t_{k-1}})^2 \mid \mathcal{F}_t \right] &= \mathbb{E}[(Y - p_t)^2 \mid \mathcal{F}_t] \\ &= p_t(1 - p_t). \end{aligned} \tag{C1}$$

Proof: expand the square of the sum. For $j < k$, condition on $\mathcal{F}_{t_{k-1}}$; the later increment has conditional mean zero, eliminating the cross term. The sum telescopes to $Y - p_t$. Boundedness supplies square integrability. Hindsight-selected turning points are not an admissible partition for this argument.

At intermediate u , the same decomposition gives

$$\begin{aligned} p_t(1 - p_t) &= \mathbb{E}[(p_u - p_t)^2 \mid \mathcal{F}_t] \\ &\quad + \mathbb{E}[p_u(1 - p_u) \mid \mathcal{F}_t]. \end{aligned} \tag{C2}$$

Remaining conditional variance decreases **in conditional expectation**, not on every path. Starting at .99, a martingale can move to .50 with probability .02 or to 1 with probability .98. On the first branch the remaining variance increases from .0099 to .25. Realized squared moves are not capped by the initial .0099.

The theorem is about a fair conditional-probability process. Observed prices contain spreads, stale information, premiums, and manipulation. A median last-hour absolute move of .50 does not verify C1 or identify a conditional standard deviation.

5.2 Release over a liquidation horizon is an additional model

Define, for $0 < p_t < 1$,

$$\begin{aligned} v_t(h) &= \mathbb{E}[(p_{(t+h) \wedge \tau} - p_t)^2 \mid \mathcal{F}_t], \\ \mathcal{R}_t(h) &= \frac{v_t(h)}{p_t(1 - p_t)} \in [0, 1]. \end{aligned} \tag{C3}$$

This definition is exact. It does not estimate $\mathcal{R}$. For $p = 0$ or 1, the ideal process is absorbed; define its future variance as zero.

The flat-release formula $v = p(1 - p)h/(T - t)$, clipped to the remaining horizon, is a particular model. It is not implied by C1. Identical current p and calendar time are compatible with no revelation during the window, immediate revelation, or many different mixtures.

An especially instructive construction: reveal all of Y at a time uniformly distributed on $[0, T]$, independent of Y . Conditional on no revelation by t , the chance of revelation in the next h is $h/(T - t)$. This **pure-jump** model has exactly the flat-release variance formula. Therefore even a correct variance release estimate does not identify Gaussian diffusion tails.

A deterministic release weighting can define $\mathcal{R} = \int_t^{t+h} \lambda(s)ds / \int_t^T \lambda(s)ds$, after truncation at T , as a model choice. Unknown future random resolution times or intensities cannot be inserted as though known at trading time. Volatility also need not increase on every realized path as the clock runs down.

5.3 Outcome-dependent information arrival

Let J denote actionable revelation during the relevant window, and set

$$\begin{aligned} a_0 &= \mathbb{P}(J, Y = 0 \mid \mathcal{F}_t), \\ a_1 &= \mathbb{P}(J, Y = 1 \mid \mathcal{F}_t), \quad \pi = a_0 + a_1. \end{aligned} \quad (C4)$$

Martingality gives

$$\mu := \mathbb{E}[p_{t+h} \mid J^c, \mathcal{F}_t] = \frac{p_t - a_1}{1 - \pi}, \quad \pi < 1. \quad (C5)$$

It does **not** give $a_0 = \pi(1 - p)$ or $a_1 = \pi p$ without outcome-independent arrival. Sharp bounds include

$$\max(0, \pi - p) \leq a_0 \leq \min(\pi, 1 - p). \quad (C6)$$

Example: from $p = .60$, early NO arrives with probability .02; otherwise probability becomes $.60/.98 = 30/49$. The next probability has mean .60. Actual adverse revelation mass is .02, whereas $\pi(1 - p) = .008$. At a 1% tail budget the incorrect independence assumption misses a full-loss atom that exceeds the entire budget.

For conditional no-revelation price variance v_d , the full fair-probability variance is

$$v_h = a_0 p^2 + a_1 (1 - p)^2 + (1 - \pi) \{v_d + (\mu - p)^2\}. \quad (C7)$$

Under independent arrival this simplifies to $v_h = \pi p(1 - p) + (1 - \pi)v_d$. Consequently, if C3 is a *total* variance estimate, the no-revelation variance must be $p(1 - p)(\mathcal{R} - \pi)/(1 - \pi)$, requiring $\mathcal{R} \geq \pi$. Adding full jump variance to an already total variance estimate double counts risk.

A bounded branch must satisfy $0 \leq v_d \leq \mu(1 - \mu)$. One optional parametric model for strict interior mean and variance is Beta with $\kappa = \mu(1 - \mu)/v_d - 1$, $\alpha = \mu\kappa$, $\beta = (1 - \mu)\kappa$. Degenerate limits need separate branches. This matches two moments, not the correct tails; empirical or robust scenario distributions may be preferable.

If arrival intensities are used, distinguish cause-specific hazards λ_0, λ_1 from variance-release weights. For deterministic conditional hazards, survival is $S(u) = \exp[-\int_t^u (\lambda_0 + \lambda_1)ds]$, and $a_y = \int_t^{t+h} S(u)\lambda_y(u)du$. Scheduled announcements require explicit probability atoms. Stochastic hazards require conditional expectations and consistency with current probability.

6. Correct tail measures and a usable margin framework

6.1 Define the loss before its quantile

For next valuation $P_h \in [0, 1]$, price-only positive loss is

$$L_h = [z(m - P_h)]^+, \quad 0 \leq L_h \leq W(z, m). \quad (D1)$$

For confidence $1 - \epsilon$, use the lower quantile convention

$$\text{VaR}_{1-\epsilon}(L) = \inf\{v : \mathbb{P}(L \leq v) \geq 1 - \epsilon\}. \quad (D2)$$

The atom-safe expected shortfall is

$$\text{ES}_{1-\epsilon}(L) = \min_{\eta \in \mathbb{R}} \left\{ \eta + \frac{1}{\epsilon} \mathbb{E}[(L - \eta)^+] \right\}. \quad (D3)$$

This general-loss formula handles probability mass at the quantile; simply taking $\mathbb{E}[L \mid L > \text{VaR}]$ does not. The exact supporting result is Theorem 10 of Rockafellar and Uryasev, *Conditional Value-at-Risk for General Loss Distributions*, with their confidence parameter $\alpha = 1 - \epsilon$.

For sorted discrete scenarios, VaR is the smallest loss whose cumulative probability reaches $1 - \epsilon$. ES averages the worst ϵ mass, using only the required fraction of a boundary atom. Round collateral upward after the calculation.

6.2 The valid endpoint-atom lemma

Theorem C: if $L \leq W$ and $\mathbb{P}(L = W) > \epsilon$, then $\text{VaR}_{1-\epsilon}(L) = W$. For every $v < W$, uncovered probability exceeds ϵ , proving the claim.

For a long the adverse revelation mass is a_0 ; for a short it is a_1 . Equality at ϵ depends on the quantile convention and other support. If loss is W with probability exactly ϵ and zero otherwise, D2 gives VaR zero.

For a Bernoulli loss, $L = W$ with probability a , zero otherwise:

$$\text{ES}_{1-\epsilon}(L) = W \min(1, a/\epsilon). \quad (\text{D4})$$

VaR jumps from zero to W when a crosses ϵ . Thus there is no universal smooth convergence to one-times leverage. A sufficiently large adverse atom forces *price-risk margin* to W under this VaR policy. It does not force every trader's actual leverage to equal one, and extra costs can require more collateral.

6.3 Compute the bounded mixture, rather than attach a jump surcharge

Let G be the conditional no-revelation price CDF on $(0, 1)$. For a long of size n and $0 \leq v < nm$:

$$\mathbb{P}(L > v) = a_0 + (1 - \pi)G((m - v/n)^-). \quad (\text{D5})$$

For a short and $0 \leq v < n(1 - m)$:

$$\mathbb{P}(L > v) = a_1 + (1 - \pi)[1 - G(m + v/n)]. \quad (\text{D6})$$

Include any additional endpoint mass from the no-revelation model explicitly. The left limit in D5 handles atoms and strict inequalities. Invert the resulting CDF, not an unrelated Gaussian approximation.

A Gaussian branch has unbounded support and contradicts bounded binary prices. Clipping introduces endpoint atoms and generally changes the mean; truncation and renormalization create another distribution. Either can be a specified approximation after recalibration, but cannot silently share the unbounded-normal derivation.

6.4 Counterexample to the proposed conservative margin

The draft's $\min(W, kn\sigma_d + \pi W + \text{execution})$ is not a general conservative VaR bound.

Set $p = m = .60$, $n = 1$, $\epsilon = .01$, independent revelation probability $\pi = .024$, $k = 3$, and zero execution cost. On the no-revelation branch, probability changes by $-.20$ with probability $.01$ or by $.20/99$ with probability $.99$. This is bounded and has mean-zero increments. Each subsequent probability can later resolve fairly. Its branch standard deviation is approximately .0201007563.

The adverse endpoint atom is $.024 \times .40 = .0096 < .01$, so an endpoint-only override does not fire. The proposed margin is approximately .0747022689. But losses exceed it with probability

$$.0096 + .976(.01) = .01936. \quad (\text{D7})$$

That is **1.936%, against the 1% target**. This counterexample preserves the draft's independent-arrival assumption; the separate example in C6 shows that assumption itself can also fail.

6.5 A genuinely distribution-free moment bound

For a zero-mean loss increment X with variance at most v , any $a > 0, c \geq 0$ satisfies

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[(X + c)^2]}{(a + c)^2} \leq \frac{v + c^2}{(a + c)^2}.$$

Minimizing at $c = v/a$ gives $\mathbb{P}(X \geq a) \leq v/(v + a^2)$. Therefore, for fair mark $m = p$, fixed size n , and a valid price variance bound v_h :

$$\text{VaR}_{1-\epsilon}(L_h) \leq \min \left(W, n \sqrt{v_h \frac{1-\epsilon}{\epsilon}} \right). \quad (\text{D8})$$

This is a conservative one-step moment bound under its hypotheses, often expensive. It does not certify v_h , identify the law of a selected informed trade, cover execution costs, or bound a running intrahorizon loss. If $m \neq p$, include the mean displacement; do not apply a mean-zero formula to a biased mark. A martingale maximal inequality can bound a running price excursion under its own conditions, but still does not create executable liquidity.

6.6 Margin policy to implement and calibrate

Model **actual closeout loss**, not only next-price movement. For a frozen position, one representation is

$$\begin{aligned} \mathcal{L}_H = & z(m - P_{\text{close}}) + \text{fees} + \text{net funding paid} \\ & + \text{other closeout costs.} \end{aligned} \quad (\text{D9})$$

Net funding paid is negative when received. Ignoring receipts would instead define a conservative loss bound. If P_{close} is the actual execution-weighted fill price, do not add the same slippage again. Multi-fill, changing-size paths need the full ledger simulation. If closure fails, follow the position to the contract's next action or terminal payout; do not drop that branch.

A candidate policy is $MM = \rho(\mathcal{L}_H^+) + K_{\text{model}}$, with a specified risk measure ρ , and $IM \geq MM + K_{\text{entry}}$, subject to explicit minimums, position caps, and uncertainty allowances. This is **policy**, requiring calibration. Statistical margin and hard capital coverage are separate tests. Trader collateral may be less than endpoint loss in the underwritten mode only while the pool covers A9/A11.

Do not cap additive fees and future charges away by forcing total margin below W . Choices are to reserve them separately, make them explicitly subordinate/payout capped, or avoid the fee-producing action when full endpoint backing is already sufficient.

Report $\mathbb{P}(\mathcal{L}_H > E)$, expected uncovered loss $\mathbb{E}[(\mathcal{L}_H - E)^+]$, VaR, ES, maximum loss, and reserve consumption separately. A 1% per-window target is not a 1% lifetime insolvency target.

7. Liquidation and compression require an executable mechanism

7.1 Trigger prices are conditional algebra

For entry equity C , absolute size n , entry e , no further charges, and maintenance fraction $0 \leq \theta < 1$ of worst price loss, solving equity equals maintenance gives

$$m_{\text{liq,long}} = \frac{e - C/n}{1 - \theta}, \quad m_{\text{liq,short}} = \frac{e + C/n - \theta}{1 - \theta}. \quad (\text{E1})$$

Apply only to an initially healthy account, and check whether the root is in $[0, 1]$. At $\theta = 1$, health is A6, independent of mark, so division is invalid. Accrued debit replaces C with current adjusted cash. State-dependent margin requires solving the actual health function; a unique root is not automatic. No trigger price guarantees a fill there after a gap.

7.2 Use the full closeout horizon

H includes data age, detection, feed publication, keeper reaction, submission, inclusion, execution queues, and all fills needed to remove risk. Grace periods add to it. Size and direction matter. News can simultaneously remove depth, increase latency, and produce the largest price jump. Empty books or chain downtime create a failure-to-close branch, potentially lasting until settlement.

A chain's nominal block interval is not a bound on H . A successful transaction that cancels or marks an account for liquidation is not a successful closeout.

7.3 Derive partial liquidation from actual execution cost

For a long closing $a \leq n$ units against executable bids $B(u)$, let

$$K(a) = \int_0^a [m - B(u)] du + \text{fees and liquidation charges.} \quad (\text{E2})$$

For a short use $\int_0^a [A(u) - m] du$ against asks. A cost upper bound must be justified if quotes can cancel. If a constant mark is used only to derive the sizing formula, the post-fill check must still use the effective current risk state.

The restoration condition is

$$\begin{aligned} E - K(a) + z_{\text{after}}(m_{\text{after}} - m) \\ \geq IM_{\text{after}}(n - a, m_{\text{after}}, \text{state}_{\text{after}}), \\ z_{\text{after}} = z - \text{sign}(z)a. \end{aligned} \quad (\text{E3})$$

The revaluation term is necessary if the mark changes during execution. For the following sizing formulas hold the mark fixed, then apply A1 and E3 to the actual final state. For a linear target $i(n - a)$, constant per-unit cost c , and initial deficit $g = in - E > 0$:

$$a_{\min} = \frac{g}{i - c}, \quad i > c, \quad a_{\min} \leq n. \quad (\text{E4})$$

The final condition is equivalent to $E \geq cn$. If it fails, even a full close at that cost leaves debt. If $i \leq c$, closing cannot repair a deficient account under this model. Add fixed cost K_0 to the numerator and per-unit fees to c . Round required quantity **up** to the lot size, then recheck exact post-fill health. An intended order quantity is not evidence of completed reduction.

For marginal linear impact, $K(a) = ca + \kappa a^2/2$, $\kappa > 0$. The requirement becomes

$$g - (i - c)a + \frac{\kappa}{2}a^2 \leq 0. \quad (\text{E5})$$

A solution requires discriminant $(i - c)^2 - 2\kappa g \geq 0$, positive improvement direction, and intersection of the root interval with $[0, n]$ and actual depth. Its lower root is

$$a_- = \frac{(i - c) - \sqrt{(i - c)^2 - 2\kappa g}}{\kappa} = \frac{2g}{(i - c) + \sqrt{(i - c)^2 - 2\kappa g}}. \quad (\text{E6})$$

The second form avoids cancellation when g is small. More closing can eventually consume too much equity under quadratic impact; do not assume “larger is always safer.” For nonlinear concentration margin, evaluate E3 directly with a justified search method.

7.4 Why independent position compression fails

The draft proposes setting remaining size to E/w to achieve one-times backing. That is not a matched trade. For two opposite one-unit positions at $m = .60$, both with equity $.20$, the proposed remaining quantities are $+1/3$ and $-1/2$, summing to $-1/6$. It has destroyed the balanced-positions invariant.

Every reduction needs an actual matched fill, funded novation, or a precisely defined paired ADL/forced-close rule. ADL at a contractual price changes users' contingent payoffs; it is not equivalent to promising the original position through resolution. A liquidator opening the opposite exposure can merely move risk rather than reduce aggregate open interest.

If a backstop inherits the entire equity $E < 0$ and exposure z , then fully backing it in isolation requires a transfer of at least $W(z, m) - E$, not merely $-E$, before extra costs. A pool may use the less conservative exact endpoint test for its total portfolio, but must test the inherited position and both sides of the transfer explicitly.

7.5 Risk phases are permissions, not proofs of capital

Use distinct states such as **trading**, **risk restricted**, **reduction pending**, **halted**, **outcome pending**, and **settled/converted**. Track verified backing separately. A scheduled deadline can restrict new risk immediately without pretending every account has already been reduced.

At a time boundary, transaction validation must derive effective permissions from time and stored rules even if no keeper called a phase-update function. Halting must remain possible with unresolved deficits; requiring everyone to become fully backed before allowing a halt can leave a dangerous market open indefinitely.

Epochs and grace periods do not make every within-epoch breach “price driven.” Funding, fees, withdrawals, new positions, oracle changes, and revised liquidity/latency inputs also change health. Their exposure must fit the funded reserve throughout the grace period.

Per-block liquidation limits can reduce immediate selling pressure but create an outstanding queue. A local feedback heuristic, such as impact coefficient times sensitivity of forced flow being below one, is conditional on a static depth and response model. It is not a global cascade-stability theorem. Simulate cancellations, simultaneous triggers, finite maker capital, and queue delay jointly.

8. Open orders are contingent obligations

8.1 Exact envelope for individually backed accounts

Suppose account (b, z) has fixed-price resting bids (n_j, P_j) and asks (a_k, A_k) , with all limit and execution prices in $[0, 1]$. Any order may fill partially, with buy prices no worse than their limits and sell prices no worse than theirs. Ignore charges momentarily.

A fill δ changes terminal equity at outcome y by $\delta(y - P)$. At NO, each bid can reduce equity and an ask improves it. At YES, each ask can reduce equity and a bid improves it. Therefore an exact robust envelope over independent fill subsets is

$$b \geq \sum_{\text{bids}} n_j P_j, \quad b + z \geq \sum_{\text{asks}} a_k (1 - A_k). \quad (\text{F1})$$

Add a conservatively derived funding and fee budget to the applicable endpoint inequalities. For genuinely independent order fills, endpoint worst cases are “all bids, no asks” and “all asks, no bids.” Partial fills are covered because the dependence is linear. Reduce-only semantics can reduce the feasible fill set, but this must be enforced at every fill, including after intervening fills change the position.

This formula is price aware and handles sign flips. Merely taking the maximum of current and hypothetical position margin at today's mark misses execution cash losses.

Draft counterexample: start with $b = 0, z = 1, m = .60, E = .60$, fully backed. Rest an ask for 2.3 units at .55. The draft's current-mark reservation compares .60 with the future short's $1.3 \times .45 = .585$, so .60 appears adequate. Execution gives $b' = 1.265, z' = -1.3$, hence YES equity $-.035$. F1 requires $b + z \geq 2.3(.45) = 1.035$ and correctly rejects it.

8.2 Underwritten books and maker validation

For leveraged resting orders, reserve the pool exposure of their permitted fills or revalidate each actual match atomically and permit rejection/cancellation of a maker that can no longer be admitted. State which promise users receive. A displayed order that is no longer funded is not guaranteed liquidity.

Every atomic match must materialize both accounts' relevant funding, apply one shared cash transfer, assess actual fees, update quantities and reserve contributions, and check the post-state. Check collateral withdrawal and replacement orders against existing reservations. Any upgrade to price, size, expiry, or risk permissions must preserve the envelope.

8.3 Order-book implementation obligations

The bounded price grid supports a compact bitmap, but it does not make matching constant cost. Work scales with fills and inspected/canceled entries. Bound both quantities, including invalid/stale makers. If a transaction exhausts its budget with a crossing remainder, do not silently rest that remainder across the spread; cancel, reject, or define a safe resumable protocol.

The draft's proposed field sizes total **248 bits**, so the listed fields can fit one 256-bit storage slot; this arithmetic is not a bug. Actual compiler layout and missing lifecycle fields still need inspection. Eight spare bits do not automatically supply expiry, risk epoch, and order-ID generation semantics. Reused IDs need generation/nonces to prevent stale cancellation and claim references.

For cumulative queue accounting, cancellations must affect the correct prefix. Example: A rests 5, then B rests 5. Cancel A, fill B's 5. A naive counter with added=10, filled=5, B starting offset=5 incorrectly gives B zero filled. Prefix cancellation rank or explicit matching is needed. Delayed user claiming must not delay the economic position or funding liability generated by a fill. The necessary accumulator is quantity weighted over the actual fill times, not just a final count.

9. Funding must preserve solvency as well as sum to zero

9.1 Units, timing, and conservation

Let r_t be collateral per claim per time, positive when longs pay; $F_t = \int r_s ds$. One possible clipped policy is

$$r_t = \text{clip} \left(\alpha(m - I) + \beta \mathbf{1}_{\mathcal{B}} \frac{m - I}{\max(\eta, \min(I, 1 - I))}, -r_{\max}, r_{\max} \right). \quad (\text{G1})$$

Here $\eta > 0$ is a boundary floor and $\mathcal{B}$ a specified boundary regime. Choose coefficient units consistently; normalized prices do not remove time units. This is an incentive policy, not a convergence proof.

Funding transfer to account i is $-\int z_i(s) r_s ds$. Since $\sum_i z_i(s) = 0$ at every economic instant,

$$\sum_i \int z_i(s) r_s ds = 0. \quad (\text{G2})$$

The correct operation order is: accrue the **old** rate to the current effective time; materialize affected accounts' old-quantity funding; execute changes; store their checkpoint; select the future rate. Cap time at feed-validity expiry or funding halt according to predefined rules. Never apply a new rate retroactively over an idle interval. Multiple transactions with identical timestamps must work with zero elapsed time.

A cumulative index alone does not make all accounting exact. Fixed-point rounding needs subunit remainder/debt accounting, and the vault must reconcile integer transfers plus fractional obligations. Independent rounding can create a credit without a corresponding debit. "Remainders exist" is not a proof that withdrawal and account closure cannot extract them.

9.2 Payer capacity

For full endpoint backing, after subtracting reserved orders and costs from slack S_i , positive funding $\Delta F \geq 0$ is allowable only if

$$\Delta F \leq \min_{z_i > 0} \frac{S_i}{z_i}. \quad (\text{G3})$$

For negative funding use shorts and $|\Delta F|$. Receiving accounts improve; paying accounts determine the bound. At exactly full backing, a payer's slack may be zero, so the allowed debit is **zero**. Example: long 100 at mark .60 with equity 60 has $b = 0$. Charging one unit of funding creates NO equity -1 without any price change.

The draft uses $(\gamma - 1)MM$ as a buffer, but its policy caps initial margin: $IM = \min(W, \gamma MM)$. Actual initial buffer is $\min(W, \gamma MM) - MM$, zero when $MM = W$. An existing healthy account may have less buffer still. Caps must use current payer slack, not an average account or an assumed initial deposit.

In underwritten mode, a funding increment changes endpoint deficits to

$$D_y(u) = \sum_i [-b_i - z_i y + z_i u]^+. \quad (\text{G4})$$

This is convex and piecewise linear. Restrict u to the interval containing zero where both endpoint requirements remain covered. For a bounded account set it can be computed exactly; more scalable implementations need a proven data structure or a conservative bound. Fee liabilities and reserve reductions must be included simultaneously.

9.3 A conservative funding bound that can be maintained cheaply

Theorem D: for unchanged balanced quantities and common increment u ,

$$D_y(b - zu, z) \leq D_y(b, z) + Q|u|. \quad (\text{G5})$$

Proof: $(x + a)^+ \leq x^+ + a^+$. Apply with $x = -b_i - z_i y$, $a = z_i u$, then sum. Balanced quantities give $\sum_i (z_i u)^+ = Q|u|$. This applies to each endpoint (indeed any fixed y) and the constant can be attained.

A sound conservative cache keeps separate upper bounds $U_0 \geq D_0$, $U_1 \geq D_1$. At each funding interval add that interval's $Q|\Delta F|$. For a nonfunding mutation touching accounts A , compute their effective pre/post deficits at the **same current funding index**, then update

$$U'_y = U_y + \sum_{i \in A} (d_{iy}^{\text{after}} - d_{iy}^{\text{before}}). \quad (\text{G6})$$

This preserves the bound, even if the local difference is negative. Pure funding materialization changes no economic balance and therefore causes no logical deficit adjustment. Before quantities change, accrue funding for the old quantities and old Q ; then update open interest.

Do not replace the segmented allowance with current Q times the absolute net index change across many intervals. Funding reversals and changed quantities can conceal historical debits. Do not subtract stale account contributions from a freshly rebased global bound. Release accumulated conservatism only after an exact reconciliation or a stronger proved decomposition. If integer implementation differs by errors ε_i in $b'_i = b_i - z_i u + \varepsilon_i$, add the adverse allowance $\sum_i (-\varepsilon_i)^+$ unless it is already covered by an exact subunit ledger.

This supplies a tractable conservative capital check, not free underwriting capacity. A crude bound may make the pool reject economically safe trades; precision and gas are an explicit tradeoff.

Future accrual must be covered before it happens. Updating the bound on the next transaction would only detect an already incurred deficit. Before enabling a rate, reserve its maximum permitted cumulative increment or impose an automatic cumulative funding cap. With cash-only reserve and other obligations already deducted, a sufficient budget for $Q > 0$ is

$$|u| \leq \frac{\min\{R - U_0, R - U_1\}}{Q}. \quad (G7)$$

Store that funding budget and an expiry; the economic accrual function must stop at exhaustion even if every keeper is offline. Quantity, reserve, order, and fee changes must revalidate outstanding budgets. For $Q = 0$, funding is zero. A feed expiry or calendar cutoff is sufficient only if the total authorized accrual up to it was actually covered. A convex funding-index envelope, described in the contract companion, is an alternative certificate with the same requirement that accrual cannot escape its funded interval.

9.4 What funding cannot prove

Requiring relative-basis pressure proportional to $|m - I| / \min(I, 1 - I)$ for every state conflicts with a uniform finite rate cap: fix $m = .5$, let $I \rightarrow 0$, and the requirement diverges. Flooring, clipping, restricting exposure, or accepting residual basis are legitimate policies.

A rule $r = \alpha(m - I)$ does not imply basis dynamics $d(m - I) = -\alpha(m - I)dt$. A response model such as expected basis drift $-\kappa r$ would instead imply half-life $\log 2 / (\kappa\alpha)$ before saturation and noise; κ needs estimation.

Long identical spot and short perp to settlement earns initial basis plus cumulative received funding, less financing/execution costs, assuming the trader can survive margin requirements. If funding later reverses, initial basis alone is not locked profit. Reverse arbitrage also requires shortability or equivalent complete-set financing. Contractual settlement at Y specifies the final accounting value; it does not guarantee a pre-halt traded price converges to Y .

10. Settlement and conversion

10.1 Fully backed conversion has an exact, mark-independent proof

Assume A6 for every account, $\sum z_i = 0$, no unreserved debits, and the ability to mint identical YES/NO complete sets at one collateral unit per pair. Give account i :

$$z_i^+ \text{ YES, } (-z_i)^+ \text{ NO, } r_i = b_i + \min(z_i, 0) \text{ cash.} \quad (H1)$$

Refunds are nonnegative. The final account payoff is

$$r_i + z_i^+ Y + (-z_i)^+ (1 - Y) = b_i + z_i Y. \quad (H2)$$

Mint Q complete sets. Total refunds are $T - Q$, so total collateral used is exactly T . This proves conservation and state-by-state payoff equality.

The mark cancels. A formula using $E(m) - W(m)$ is just A5 in disguise. A malicious conversion mark cannot affect pure, consistent full-size conversion. A preceding forced close at that price, inconsistent equity rebasing, or different token semantics can affect payouts; those are separate operations requiring their own protections.

Current Polymarket documentation describes paired outcome tokens backed by one unit of its specified collateral. That is a mechanism reference, not proof that a Monad contract can immediately redeem or mint the identical asset. Cross-chain collateral, redemption, and event mappings must be separately verified. [Polymarket position mechanics](#).

10.2 Extension: tokenize insured positive claims directly

Exact pooled coverage A9 need not supply enough cash to top up *every* account to individual backing at once. There is nevertheless another conversion, if the promised product is limited-liability positive terminal equity.

This extension is exact only for strictly binary final outcomes. Set $c_{i0} = [b_i]^+$, $c_{i1} = [b_i + z_i]^+$, and $M = \max(\sum_i c_{i0}, \sum_i c_{i1}) = T + R_*$. Mint M complete sets. Give each account c_{i0} NO tokens and c_{i1} YES tokens. Its payoff at either binary endpoint is exactly its promised positive terminal equity. The reserve receives leftover tokens on each side and any unused cash.

If available reserve is $R \geq R_*$, the reserve's final remaining assets equal $R - D_Y$. Proof: unassigned outcome tokens are $M - P_Y$; unused cash is $T + R - M$; their sum is $R - D_Y$ by A8. This proves both funding sufficiency and residual reserve accounting.

This differs from H1: it tokenizes the **insured payoff**, including the default option, rather than preserving original position size plus a nonnegative cash refund. At a fractional invalid payout y , these tokens pay the chord $(1 - y)[b_i]^+ + y[b_i + z_i]^+$, which can exceed $[b_i + z_i y]^+$. For $b = -480$, $z = 1000$, $y = .5$, it pays 260 instead of 20. Preserve cash settlement for such cases or specify that changed invalid payoff before trading; do not claim equivalence. H1's linear conversion remains valid for matching fractional token payouts. Granularity, fees, timing, and identical token semantics remain requirements. Conversion allocates funded losses; it does not make the insurance free.

10.3 Cash settlement and an explicit shortfall rule

Cash settlement is directly $v_i = b_i + z_i Y$ after funding and charges stop according to the specified cutoff. Using $E(m) + z(Y - m)$ is equivalent only when both terms use the same mark and accrual snapshot. Adding a terminal adjustment to stale or inconsistently marked equity double counts or misses P&L.

In individually backed mode, a negative v_i is an invariant failure. In underwritten mode, pay v_i^+ , consume actual deficit D_Y , and record defaults exactly. Do not silently switch modes when the code encounters a negative number.

If residual available assets $A \geq 0$ are less than positive claims $P = \sum_i v_i^+$, one coherent disclosed loss-sharing rule is

$$\alpha = \min(1, A/P), \quad \text{payout}_i = \alpha v_i^+, \quad P > 0. \quad (\text{H3})$$

Handle $P = 0$ separately. Specify senior liabilities before defining A . “Only haircut winners' profits” is another possible rule, but it may lack enough haircut capacity; derive and disclose its fallback. ADL ranking, close price, and which payoffs are extinguished must similarly be fully specified.

Compute denominators and loss factors from a frozen complete snapshot. Batch processing must not permit withdrawals by early claimants before total liabilities are known. Freeze the account enumeration; a swap-and-pop list changing under a settlement cursor can skip accounts. Payout rounding should never exceed assets; dust ownership and duplicate-claim prevention must be explicit. Aggregate integer invariants require their own proof beyond the real-number H3 formula.

Invalid, disputed, canceled, delayed, and unresolvable events require ex ante terms. A .50 refund, last TWAP, and voiding trades are different payoffs. A source can permit fractional payouts; coercing those to YES or NO changes the instrument.

11. Oracle, mark, manipulation, and collateral risk

11.1 Three different oracle jobs

Separate **risk valuation**, **actionable-information monitoring**, and **final resolution**. A monitor can restrict new risk when evidence appears; that does not mean the outcome is final enough to release assets. A correct final outcome does not establish that the earlier index was fresh or liquidations were fairly priced.

For every observation, authenticate event identity, source, signer, sequence, observation time, publication time, accepted time, range, and revision policy. Specify disagreement, staleness, missing data, reorg, and downtime behavior. A signed wrong value is still wrong. Freezing a stale mark can make reported margin look stable while actual risk grows.

The standard microprice expression $(P_{ask}Q_{bid} + P_{bid}Q_{ask})/(Q_{bid} + Q_{ask})$ is an imbalance heuristic, not a guaranteed fair value. Midprice, microprice, and TWAP from one book are three transformations of one corruption domain. Taking their median is not equivalent to independent sources.

TWAP must accrue the *previous* observation over its actual elapsed duration, handle gaps, and avoid treating unavailable data as genuine persistence. Per-update clamps can be bypassed by many small updates unless movement is also bounded over a fixed time window. Conversely, clamping a real jump can keep the engine trading against stale risk. The response must be risk restriction backed by capital, not a false declaration that price could not move.

11.2 Manipulation economics need a threat model

Under a deliberately simple fixed book with depth d per price step δ , sweeping N adverse levels at distances $\delta, 2\delta, \dots, N\delta$ costs slippage

$$C_{\text{sweep}} = d\delta \frac{N(N+1)}{2} \approx \frac{d\Delta^2}{2\delta}, \quad \Delta = N\delta. \quad (I1)$$

This is not a universal attack-cost lower bound. Attackers may own quotes, cancel rather than sweep them, wash trade, hedge externally, or profit from liquidation, funding, rebates, reserves, and linked markets. Sustaining a TWAP manipulation depends on replenishment and adversarial market response; an assumed refill rate is an input, not security evidence.

For a stated feasible strategy set $\mathcal{A}$, the desired economic claim is

$$\sup_{a \in \mathcal{A}} \{ \mathbb{E}[G(a)] - C_{\text{irreversible}}(a) - C_{\text{carry}}(a) - \mathbb{E}[\text{collectible penalties}(a)] \} \leq 0. \quad (I2)$$

Any practical finding is conditional on coverage of that strategy set and the cost/gain estimates. Open-interest limits should constrain stake exposed to each source/corruption domain, not rely only on gross public volume. Test coalitions across wallets; self-trade prevention is insufficient.

11.3 AI agreement does not prove settlement accuracy

If optional AI evidence agents have equal error variances, the variance-based effective number of equally weighted observations is

$$n_{\text{eff}} = \frac{n^2}{\sum_{ij} \rho_{ij}} = \frac{n}{1 + (n-1)\rho} \quad (I3)$$

for common pairwise correlation.

This concerns variance of an average, not majority-vote accuracy. Three exchangeable voters can each be .82 accurate with pairwise correctness correlation .60 under either of these distributions for number correct K :

K	Distribution A	Distribution B
0	.12096	.06192
1	0	.17712
2	.17712	0
3	.70192	.76096

Both have the same individual accuracy, pairwise correlation, and n_{eff} . Majority accuracy is **.87904** versus **.76096**. Thus those summary statistics cannot certify a confidence gate.

Combining individual posteriors c_i by Bayes requires conditional independence of the underlying evidence given Y and a common prior π_0 :

$$\begin{aligned} \text{logit } \mathbb{P}(Y = 1 \mid E_1, \dots, E_n) \\ = \sum_i \text{logit } c_i - (n - 1) \text{logit } \pi_0. \end{aligned} \quad (\text{I4})$$

Calibration of individual scores alone does not imply that factorization. Shared retrieval, shared model training, copied evidence, and ambiguous rules can produce correlated failure. Shrinking log-odds by an effective-sample-size ratio is a heuristic requiring a held-out test.

The cited oracle experiment's 97.87% cell covers 563 unanimous/high-confidence cases selected by a post-hoc gate. It does not establish production certainty or the accuracy of a newly chosen voter configuration. Use a frozen gate, independent event-cluster evaluation, and monetary loss weighting. [Kota, exact v1 study](#).

Even under independent identically distributed trials, zero observed errors in N cases has one-sided 95% upper error bound $1 - .05^{1/N}$. To push that bound below 10^{-4} takes 29,956 zero-error cases. Dependence, gate selection, and distribution shift make that calculation inapplicable without further work. No such validation was performed here.

11.4 Resolution stake, bonds, and escalation

For a fully signed, balanced book, changing settlement from y to y' reallocates gross stake

$$\frac{1}{2} \sum_i |z_i(y - y')| = Q|y - y'|. \quad (\text{I5})$$

Limited liability and reserves require computing the actual payout change; linked positions and attacker external gains can enlarge the attack incentive. Do not use the largest single account as the coalition's maximum gain.

If success produces gain G , a bond B is forfeited with probability P_f , and attack cost is C , deterrence requires

$$P_s G - P_f B - C \leq 0. \quad (\text{I6})$$

The familiar $(1 - P_d)G - P_d B$ is a special case with complementary success and detection/slashing probabilities. Detection is not necessarily slashing; challenge capital constraints, ambiguous adjudication, and a correlated watcher outage matter. A common outage probability u with otherwise independent watcher failures f_i gives no-challenge probability $u + (1 - u) \prod_i f_i$, not merely $\prod_i f_i$.

Deferral should compare the expected reduction in mispayment, roughly $(p_{\text{auto}} - p_{\text{deferred}}) \times \text{loss severity}$, against extra locked-capital carry, review costs, and grieving effects. Deferred resolution is not necessarily error free, and the capital locked is not always just open interest. Outcome truth, bond availability, adjudication, and payout liquidity need separate stress cases.

11.5 Collateral and operational assumptions belong in the claim

Cash-token solvency does not imply dollar solvency after depeg. Unsupported transfer fees, rebasing, frozen addresses, bridge losses, inaccessible redemption, compromised upgrade keys, and privileged outcome changes can invalidate the asset assumptions. Whitelist a precisely specified collateral behavior and verify actual received amounts. If reserve capital is invested, haircut it by enforceable stressed realizable value and availability, not nominal balance.

For Monad deployment, pin and measure actual transaction gas limits, ordering, timestamps, execution verification, and finality behavior. The chain's performance does not remove jump losses or guarantee keeper inclusion. There is no deployment or integration verification in this report. Current [Monad developer essentials](#) are an integration reference to recheck against the selected network and software versions.

12. Repairing the draft's worked example

Retain its illustrative assumptions: mark .66, one-hour window, flat release, $k = 3$, half-spread .005, initial multiplier 1.5, no modeled arrival jump, no impact, and

$$i(\Delta) = \min\{w, 1.5[3\sqrt{.66(.34)/\Delta} + .005]\},$$

with Δ in hours. This is a recalculation of the draft's model, not endorsement of it as conservative margin.

At 48 hours:

$$\sigma = .0683739716559, \quad i = .3151828724515.$$

The long leverage cap is approximately 2.09402; the short cap is 1.07874. Alice, long 1,000 with equity 250, needs at least **65.1828724515** additional collateral; 65 is insufficient. For an illustrative cents-level top-up use 65.19. Bob, short 1,000 with equity 150, needs to buy back

$$a_{\min} = \frac{1000i - 150}{i - .005} = 532.5338280157.$$

If this example uses whole-claim lots, close **533**, not 532. With a .001-claim lot convention, the rounded requirement would be 532.534 instead; state which convention an example uses. Closing 532 leaves equity 147.34 against required 147.5055843.

At 12 hours, the formula gives long margin .6228657449 and long leverage **1.0596183935**, not one. Solving for when the long first reaches its full .66 margin gives

$$\Delta_* = \frac{k^2 p(1-p)h}{(w/\gamma - s)^2} = 10.6730083234 \text{ hours}, \quad (\text{J1})$$

when $w/\gamma > s$. If that denominator condition fails, spread alone already hits the policy's cap. A mandatory full-backing floor at 12 hours is a permissible *additional policy*, but is not derived from this example's formula.

Here is one complete repair using whole-claim lots and an explicit 12-hour full-backing floor:

- Alice tops up 65.19 at the first check, then another 344.81, reaching equity 660 at the unchanged .66 mark.
- Bob buys 533 at .665, pays 2.665 spread, retains short 467 and equity 147.335; a later 12 top-up gives 159.335.
- A maker with deposited equity 200 sells those 533 at .665, retaining short 533 and equity 202.665 at .66.

Final quantities sum to zero, deposits total 1,022, and the canonical ledger gives:

Account	Position	Equity at .66	NO payout	YES payout
Alice	+1,000	660	0	1,000
Bob	-467	159.335	467.555	.555
Maker	-533	202.665	554.445	21.445
Total	0	1,022	1,022	1,022

Full-size conversion mints 1,000 complete sets and refunds cash 0, .555, and 21.445. This is exactly funded. This repaired trace excludes funding and additional fees; a separate .67 funding-basis illustration in the draft cannot be inserted without recomputing the account ledger and ensuing requirements.

13. Claim-by-claim verdict on the attached draft

Page numbers refer to the 38-page theory/mechanism PDF. “Conditional” means the statement needs the written restriction, not that the implementation has met it.

Location / claim	Verdict	Required correction or supporting result
§2.1, p.2: conditional probability is a martingale	Proved	Increasing filtration, binary integrable outcome; distinguish price measures
§2.1: discounted physical probability is the traded price	Conditional	Pricing measure, discounting, payment date, friction assumptions
§§2.3-2.5: worst loss and entry accounting	Proved with scope	A1-A6; cash ledger, sign flips, pending funding and fees
§3.1, pp.4-5: expected squared moves equal $p(1-p)$	Proved with sampling restriction	C1; nonanticipating partition and optional sampling
§3.2: variance budget cannot grow	False pathwise	C2; .99→.50 counterexample
§3.2: median final-hour move verifies conditional volatility	Not established	Different statistic and conditioning; observed prices need calibration
§3.3 / §5.5: flat release follows from variance identity	False as derivation	C3 is a separate model; pure-jump model shares its variance
§4.2: liquidation-price formulas	Conditional	E1; $\theta < 1$, initial health, fixed charges, no fill guarantee
§4.3: surviving leveraged account has adverse endpoint deficit	Proved	B1; stated no-intervention assumptions
§4.4: protected terminal payoff equals fewer spot tokens	Conditional identity	B2-B3; fixed charges and identical claims; no “leverage has no value” generalization
§5.2: arrival outcome probabilities follow from martingality	False	C4-C7; directional joint probabilities
§5.3: adverse endpoint atom above tail budget forces full VaR	Proved after correction	Theorem C; strict inequality, bounded loss, correct atom
§5.3: Gaussian margin continuously approaches bounded W	Not established / inconsistent	Unbounded support; atom discontinuity in D4
§5.4: additive $k\sigma + \pi W$ rule is conservative	False in stated generality	1.936% exceedance counterexample D7
§5.4: “nontrivial” arrival forces actual leverage exactly one	Overstated	Threshold depends on side, tail budget, convention; costs may require more capital
§6.1: only price causes intra-epoch breaches	False	Funding, fees, withdrawals, fills and risk-input changes
§6.2: partial-close sizing	Conditional	E3-E6; actual execution, revaluation, feasibility, upward rounding
§6.3: independent $x' = E/w$ compression guarantees backing	False as a mechanism	Unbalanced residual positions; matched action or explicit payoff change required
§6.4: full-size backed conversion conserves value	Proved	H1-H2; identical claims and available collateral
§6.4: pure conversion requires a trustworthy halt mark	Unnecessary under correct ledger	Mark cancels; prior forced trades are separate

Location / claim	Verdict	Required correction or supporting result
§7.1: funding coefficient supplies a price half-life	Conditional model, not theorem	Need measured price-response coefficient
§7.2: spot/perp arbitrage locks basis with future funding	False generally	Include cumulative funding, financing, survivability, shortability
§7.3: lazy funding is zero-sum	Conditional	Pathwise G2 plus rounding, position-time and cache proofs
§7.4: finite funding cannot control relative basis uniformly at boundaries	Proved with scope	Fixed nonzero absolute basis as index approaches boundary
§7.5: $(\gamma - 1)MM$ is available payer buffer	False under capped IM	Current slack; G3-G7; funding may need to stop
§8: robust mark formulas establish manipulation resistance	Incomplete	Independent corruption domains, strategy/cost bounds, stale-news behavior
§9: static depth/cascade inequality establishes stability	Conditional heuristic	Endogenous cancellations, finite capital and queue simulation
§10: bounded grid permits a compact bitmap	Valid data-structure direction	Matching still has bounded inspection/fill obligations
§10: proposed order fields fit a slot	Arithmetic valid: 248 bits	Verify compiler layout and lifecycle fields; no fabricated 288-bit objection
§§10-11: cumulative fill accounting handles cancellation/funding	Incomplete	Prefix cancellation and economic timing of deferred fills
§11: two hypothetical position-margin maxima reserve all orders	False	.035 deficit example; exact price-aware F1 envelope
§12: cash terminal settlement removes need for final trading convergence	Correct accounting statement	Consistent snapshot and funded/default-mode claims
§13: effective voters plus confidence establish resolution accuracy	Not established	I3-I4 nonidentification, calibration and coalition stakes
§14, pp.34-36: numerical example follows its stated rules	Incorrect	533 whole units; corrected top-ups and 10.673-hour threshold
§15: these components guarantee zero protocol bad debt	Not proved; false without additional backing rules	Need initial coverage and preservation under every admitted transition

The largest issue is not a missing decimal. It is the unproved transition from “risk requirements rise” to “funded counterparties or collateral arrive before revelation.”

14. The actual safety theorem to implement

Conditional engine theorem. Fix the payout/default model, asset custody assumptions, allowed outcome set, and fee/funding seniority. Suppose:

1. The initial state satisfies custody reconciliation, matched positions, and the chosen coverage certificate.
2. Every permitted atomic transition preserves those invariants, including reservations for all obligations it authorizes to accrue later.
3. Time-dependent obligations stop at enforceable funded limits without depending on a keeper acting on time.
4. Settlement uses the declared immutable outcome and an order-independent, correctly rounded allocation of the same claims and assets.

5. External assumptions used in the certificate actually hold: token behavior, custody, hedge equivalence and availability, and outcome semantics.

Then, by induction over transitions and the bounded-accrual condition between them, all promised terminal claims in the declared scenario set can be paid from the secured assets. The settlement step follows A8-A11 or the deliberately limited payout rule. This is the theorem worth refining into code verification.

It proves payment under a model of assets and transitions. It does not prove oracle truth, insurer profitability, stablecoin dollar value, available liquidation depth, or unrestricted chain liveness. Those need separate evidence or explicitly limited promises.

A concrete implementation sequence

1. **Fix semantics:** payout, invalid events, capital mode, loss waterfall, settlement clock, reserve rights, and early termination permissions.
2. **Build an exact reference engine:** A1/A2 plus actual fees, order reservations, funding budgets, reserve updates, lifecycle, and settlement. Encode both adversarial outcomes from the start.
3. **Write the contract around those transitions:** vault accounting, position ledger, bounded matching, risk certificate, funding accumulator, lifecycle, resolution adapter, and claims. Separation into contracts is an engineering choice; atomicity of economic operations is mandatory.
4. **Refine rational arithmetic to integers:** compatible units, directed bounds, bounded products, carries, and identical cash transfer on both trade legs.
5. **Prove/test the implementation:** property and state-machine tests, differential reference comparison, small exhaustive domains, formal arithmetic/transition proofs, and external security review of actual code.
6. **Calibrate economic policy:** causal data, event-cluster holdouts, actual depth/latency, finite underwriting, informed flow, and market-maker behavior.
7. **Validate real integrations:** exact source/event mapping, collateral behavior, finality, failure/recovery paths, gas and processing limits.

For example, with six-decimal collateral, a .001 price tick and .001-claim lot produce exactly one collateral atom per lot per tick. Trade cash is integer $n_{\text{lots}} k_{\text{tick}}$, and terminal payout per lot is 1,000 atoms. This is a compatible illustrative convention, not a production parameter recommendation. Funding and fees still need their own scales. Solidity signed division rounds toward zero; arithmetic conventions must be intentional and tested. [Solidity integer arithmetic documentation](#).

The [contract companion](#) supplies the operation matrix, 20 named invariants, bookkeeping for lazy funding, settlement modes, and required tests. Those requirements are not represented as completed contract tests.

15. What risk analysis must measure

The [validation companion](#) provides the detailed data schema, parameter dictionary, simulation layers, and release gates. At minimum:

Risk	Required input/model	Required output / failure measurement
Terminal default	Canonical accounts, allowed outcomes, secured assets	Endpoint deficits, reserve consumption, residual unpaid claims
Information jumps	Outcome-linked announcement timing, directional arrival, private information	Conditional loss tails, adverse selection after trade acceptance
Liquidation execution	Depth changes/cancellations, queue priority, actual fills, end-to-end delay	Completion fraction, residual exposure, failed-close rate, cost distribution

Risk	Required input/model	Required output / failure measurement
Funding	Actual position-time paths, rates/caps, payer budgets, idle periods	Cash/remainder conservation, budget exhaustion, reserve exposure
Margin policy	Frozen parameter version, uncertainty and cost assumptions	VaR exceedances with intervals, ES, forced reductions and recovery opportunity cost
Underwriting	Finite initial reserve, premiums, hedges, costs, withdrawals	Lifetime ruin probability, loss severity, return on committed capital
Manipulation	Coalition strategy set, irreversible costs, cross-market gains	Profitable attack traces, stake per corruption domain, limit sensitivity
Resolution	Rule ambiguity, disputes, source dependence, watcher incentives	Monetary misallocation, finality delay, wrong-outcome and no-challenge risk
Custody/integration	Actual token/bridge/oracle/role behavior	Asset unavailability, depeg exposure, privileges, recovery correctness
Contract execution	Compiled implementation, finite types/work bounds, target chain	Invariant failures, gas/queue tails, reverts, duplicate or unfair claims

Use exact accounting tests, mechanical historical replay, and behavioral/adversarial simulation as three distinct layers. Do not infer equilibrium from replaying a fixed historical path. Do not infer actual fill availability from candles. Do not condition away early revelation, canceled markets, missing depth, or failed execution.

For each model and policy version, record: training/calibration/test periods; event-cluster split; source hashes; estimator; confidence interval; stressed alternatives; default for missing inputs; maximum authorized parameter range; and who may change it. Values such as $5\times$ leverage, one-hour closeout, 99% confidence, a particular fee, or a 12-hour restriction remain illustrative until supported by that record.

Useful capital statistics include residual loss $S = [D_Y - R_{\text{usable}}]^+$, $\mathbb{E}[S]$, tail measures of S , and lifetime reserve ruin probability. Track reserve cashflows explicitly:

$$\begin{aligned}
 R_{k+1} = & R_k + \text{earned premiums} + \text{allocated fees} \\
 & + \text{explicit injections} - \text{losses paid} \\
 & - \text{costs} - \text{withdrawals}.
 \end{aligned} \tag{K1}$$

This cashflow equation must be reconciled to the full custody ledger and any changing hedge assets. Do not count customer collateral again as insurance, assume capital resets between stress events, or hide injections as fee income. Report integrated uncovered exposure over time, not only the final balance, because a later top-up does not mean an earlier full-payment promise was funded.

Any launch risk appetite - acceptable loss frequency, reserve return, maximum oracle exposure, or operational delay - is a policy decision. Data can estimate consequences; it cannot decide the user's acceptable risk. No production values for those tolerances are invented here.

16. Evidence delivered and limits

This analysis distinguishes four levels of evidence:

- **Algebraic proofs:** the ledger, endpoint collateral, exact reserve, conversion under its stated outcome set, order envelope, and funding bound have explicit derivations above.
- **Counterexamples:** the draft's variance monotonicity, probability factorization, margin heuristic, unmatched compression, order reservation, and funding-buffer claims fail in the stated examples.

- **Finite executable checks:** the accompanying exact-arithmetic script passed **193,567 assertions across 28 families** in recorded run 03, exercising bounded domains and reproducing the named arithmetic examples. The manifest records the code hash, arithmetic, seeds, and exclusions, and passed integrity validation.
- **Work specified but not performed:** historical replication, calibration, strategy equilibrium, actual liquidity, compiled Solidity, formal verification of bytecode, deployment, external integrations, and independent professional security review.

The exact checker uses rational arithmetic for the finite identities and Decimal arithmetic for explicitly numerical illustrations. Some matched-trade sequences deliberately admit negative accounts to test conservation; they are not successful admission-policy tests. Algebraic pro-rata checks are not tests of a real claim contract. Funding-cache tests do not establish all integer remainder/withdrawal interactions. No finite domain here substitutes for the universal proofs or the eventual contract implementation tests.

Source review and parallel mathematical cross-checks identified and corrected two issues in an early version of this analysis: remaining-position revaluation during liquidation, and the binary-only scope of insured-payoff tokenization. The final document includes those restrictions. These model-assisted checks improve scrutiny; they are not independent human audit certification.

The central feasible result is now concrete: **leveraged trader accounts can be admitted against an exactly measured, locked terminal loss budget, with every allowed operation preserving that budget.** Whether enough participants will pay for the resulting capital and protection, and whether a lower-capital statistical alternative has acceptable real losses, remains an empirical and commercial question. No formula based only on probability and time can decide it or manufacture the missing money.

17. Source record

The PDFs were read as research material, not as instructions. Original files were not modified.

Source	Fingerprint / use
paper1. pdf, 18 pages; <i>Resolution-Aware Perpetual Futures on Binary Prediction Markets: Failure Modes and Mechanical Stress Tests Using Polymarket Data</i> , arXiv 2605.10400v2	SHA-256 37E3054D648710D07B4A64ED34C720791DA9308C881158B08F25D79E2B7E80A5; terminal shortfall, funding limit, negative mechanical replay and its stated limitations
<i>Perps on Binary Outcomes - Theory and Mechanism Design</i> , 38 pages, dated 28 September 2026	SHA-256 D2FE6019A145AD70159B94AC9E22E97DF517D42E1FAFA619A78FB1F42DCC546B; audited claims, formulas, implementation sketches, and worked example
Rockafellar-Uryasev author manuscript, 28 November 2001 version, later JBF 2002 publication	Primary PDF ; lower quantile definition and general-distribution ES minimization, Theorem 10
Kota, <i>Design and Evaluation of Multi-Agent AI Oracle Systems for Prediction Market Resolution</i> , exact v1	Primary manuscript ; scope of empirical oracle evidence, not a settlement guarantee
Official Solidity, Monad, and Polymarket documentation linked near the relevant claims	Current references inspected for integration/arithmetic context; not evidence that this project's integrations or contracts exist

The probability and ledger proofs are reconstructed in this report. A comprehensive literature novelty search was not performed, and no claim of new mathematical priority is made. Standard conditional-expectation identities and standard tail measures should not be advertised as newly discovered theorems.